

## About

---

I am a **Ph.D. student** in the **Graduate School of Culture Technology at KAIST** and a member of the **Ubiquitous Virtual Reality Lab**, advised by **Prof. Woontack Woo**. I believe **physical space can become a new medium for AR/VR/MR**, and my work explores how **spatial computing** can turn heterogeneous environments into shared, collaborative experiences.

### RESEARCH INTERESTS

Augmented Reality · Ubiquitous Virtual Reality · Remote Collaboration · Spatial Computing

## Publications

---

### 2026

#### **Spatial Affordance-Aware Affine Transformation Between Heterogeneous Spaces for Mixed Reality Remote Collaboration**

Seonji Kim, Dooyoung Kim, Selin Choi, Woontack Woo  
IEEE TVCG 32(5), 2026

#### **Sharing Augmented Reality Furniture across Heterogeneous Spaces: Asymmetric Effects of Unmatched Area Visualization**

Seonji Kim, Jinwook Kim, Selin Choi, Minju Baeck, Boram Yoon, Jens Emil Grønbaek, Jae-eun Shin, Woontack Woo  
TVCG submission, 2026

#### **Task Breakpoint Generation using Origin-Centric Graph in Virtual Reality Recordings for Adaptive Playback**

Selin Choi, Dooyoung Kim, Taewook Ha, Seonji Kim, Woontack Woo  
IEEE TVCG 32(5), 2026

#### **VR Zen Garden: Designing a Virtual Environment for Stress Relief**

Hail Song, Jinseok Hong, Seonji Kim, Kyung Taek Oh, Woontack Woo, Sungyoung Kim  
IEEE VR Workshops, 2026

### 2025

#### **Architectural Elements Augmentation from 2D Artworks for Spatial Experience of AR Exhibition**

Young Bin Kim, Suji Kang, Minju Baeck, Seonji Kim, Woontack Woo  
IEEE ISMAR Adjunct, 2025

#### **Human-Scene Interaction Data Generation with Virtual Environment using User-Centric Scene Graph**

Taewook Ha, Selin Choi, Seonji Kim, Dooyoung Kim, Woontack Woo  
IEEE VR Workshops, 2025

#### **Meta-Objects: Interactive and Multisensory Virtual Objects Learned From the Real World for Use in Augmented Reality**

Dooyoung Kim, Taewook Ha, Jinseok Hong, Seonji Kim, Selin Choi, Heejeong Ko, Woontack Woo  
IEEE Computer Graphics and Applications 45(3), 2025

#### **Mutual Space Representation Standardization for Mixed and Augmented Reality Remote Collaboration**

Seonji Kim, Woontack Woo  
IEEE ISMAR Adjunct, 2025

---

## 2024

### Object Cluster Registration of Dissimilar Rooms Using Geometric Spatial Affordance Graph to Generate Shared Virtual Spaces

Seonji Kim, Dooyoung Kim, Jae-eun Shin, Woontack Woo  
IEEE VR, 2024

### Spatial Affordance-aware Interactable Subspace Allocation for Mixed Reality Telepresence

Dooyoung Kim, Seonji Kim, Selin Choi, Woontack Woo  
IEEE ISMAR, 2024

## 2023

### The Effects of Spatial Configuration on Relative Translation Gain Thresholds in Redirected Walking

Dooyoung Kim, Seonji Kim, Jae-eun Shin, Boram Yoon, Jinwook Kim, Jeongmi Lee, Woontack Woo  
Virtual Reality 27(2), 2023

## 2022

### Bring Store in My Room: AR Store Authoring System for Spatial Experience in Mobile Shopping

Seonji Kim, Hyuckjin Jang, Kyung Taek Oh, Seoyoung Oh, Dooyoung Kim, Woontack Woo, Jeongmi Lee, Jaehong Ahn, Sang Ho Yoon  
IEEE ISMAR Adjunct, 2022

### TranSlider: User-Controlled Augmented Content Presence Visualization in AR Collaboration

Boram Yoon, Seonji Kim, Woontack Woo  
IEEE ISMAR Adjunct, 2022

---

## Research Experience

|              |                                                                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024–Present | <b>Meta-Object</b><br>Real-time XR Interface Technology Development for Environmental Adaptation<br>Intelligent Platform Team · IITP, Ministry of Science and ICT |
| 2021–2025    | <b>OpenXR: TranSpace 3.0</b><br>Development of Untact Realistic OpenXR Platform Technology<br>Real-World Digital Twin Generation and Management Team              |

---

## Education

|              |                                                                           |
|--------------|---------------------------------------------------------------------------|
| 2023–Present | <b>Ph.D. Student · KAIST</b><br>Graduate School of Culture Technology     |
| 2021–2023    | <b>M.S. · KAIST</b><br>Graduate School of Culture Technology              |
| 2016–2021    | <b>B.S. · Sungkyunkwan University</b><br>Computer Science and Engineering |

---

## Teaching

|        |                                                                    |
|--------|--------------------------------------------------------------------|
| GCT445 | <b>Augmented Reality</b><br>Teaching Assistant · Summer 2025       |
| GCT600 | <b>Augmented Reality Project</b><br>Teaching Assistant · Fall 2024 |

---

|               |                                                                    |
|---------------|--------------------------------------------------------------------|
| <b>GCT555</b> | <b>3D Interaction Design</b><br>Teaching Assistant · Spring 2024   |
| <b>GCT600</b> | <b>Augmented Reality Project</b><br>Teaching Assistant · Fall 2023 |